Workshop Parts Desk

A safe, budget-aware AI agent for a parts counter

Team F60 - HackMatrix 1.0 (KSIT) - Agentic AI

100 / 100 official, before and after the twist - 0 unsafe dispatches

WHAT WE HAD TO SOLVE

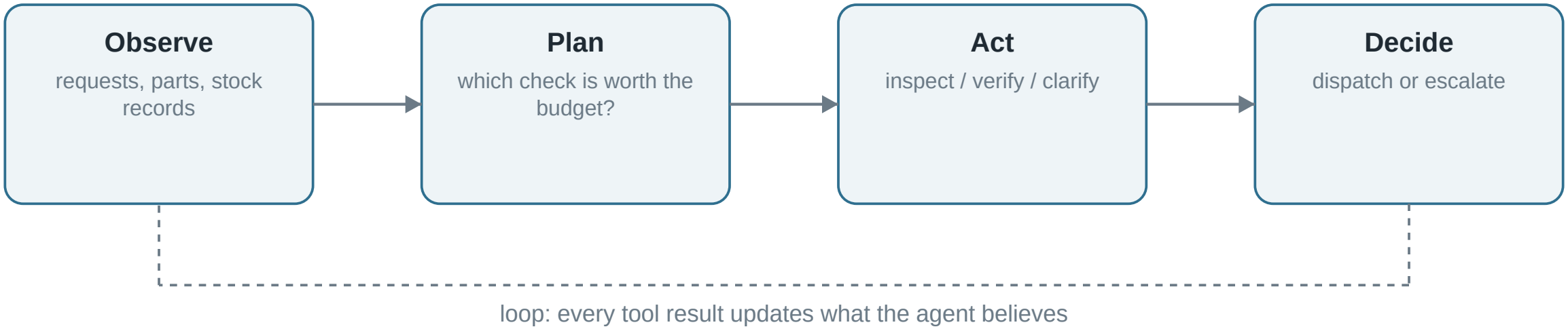
The problem

- 6 customer requests in everyday language + 12 power supplies described in text
- For each request: dispatch a matching part, or escalate to a human
- Match = same voltage + same connector + watts inside the allowed range
- Only 14 tool units for the whole batch (inspect 1, verify 2, clarify 2)
- ONE wrong part = the whole batch scores ZERO

Traps: stale drafts, mV/mW units, aliases, fake numbers, out-of-date stock

NOT A CHATBOT - IT ACTS

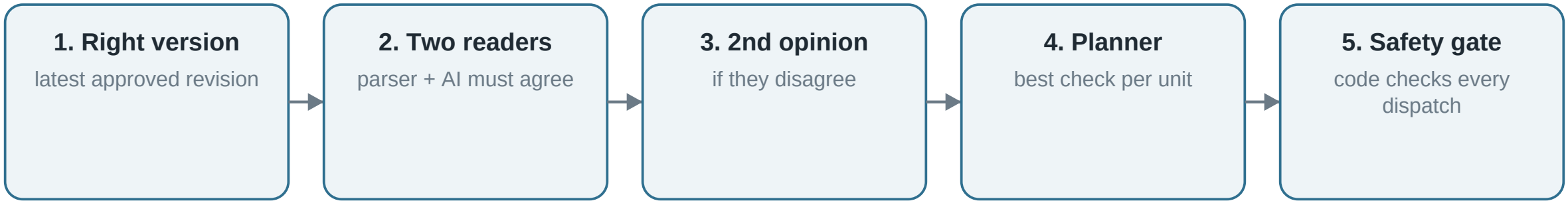
Why it is agentic



- Makes irreversible decisions on its own, under a hard budget

FIVE STEPS

How it works



- The AI reads the text; plain code makes the decisions
- Deterministic: same input, same decisions, every time

0 UNSAFE DISPATCHES - EVER

Safety first

- The AI never decides alone: the AI and a rule-based parser must agree
- Part specs are confirmed by the official tool card before use
- Unknown watt range? Ask the customer first - never guess
- Final code check before every dispatch: voltage, connector, range
- Tested against a deliberately wrong AI: still 0 unsafe

0 unsafe dispatches in every test and every official run

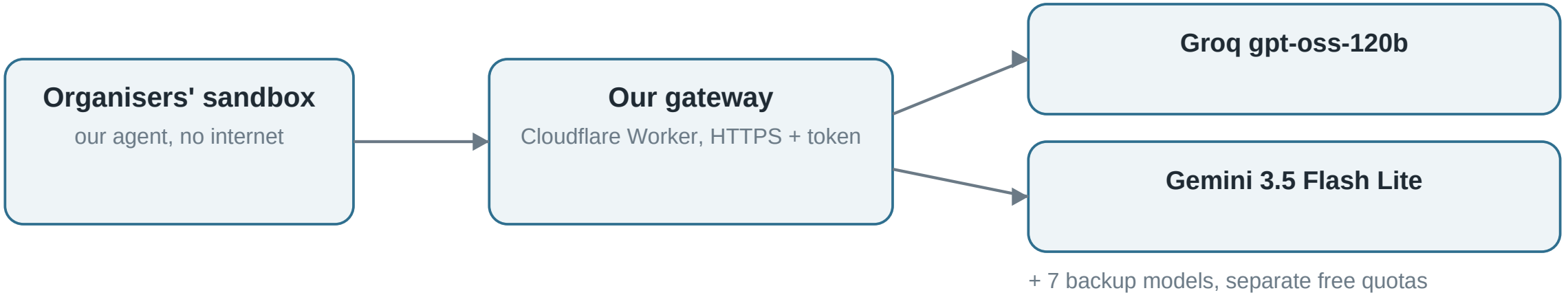
EXPECTED-VALUE PLANNING

Smart use of the budget

- For each possible check: how much does it raise the expected score per unit?
- Pick the best check, read the result, repeat
- Cheap exact check (inspect) when the desk is up to date; verify only when needed
- Hard stop at 13 units - keeps the full efficiency bonus
- Average spend: 11-12 units of 14

HOW THE AGENT REACHES ITS AI

Our AI gateway on Cloudflare



- Free, always on, HTTPS on port 443, keys stored as encrypted secrets
- Races two models; falls back across 9 if one is busy - answers in 2-5 s

NEW RULES MID-HACKATHON

Adapting to the twist

CHANGE	OUR FIX
Watts became a min-max range	match inside the range only
"Contact the customer for limits"	clarify first, read the full reply
Stock came as documents + manifest	read the exact quantity
Desk behind the warehouse	skip inspect, verify directly
Shared stock (2+ units per part)	one part can serve several requests

Old code re-run: 51 / 39 -> final version: 100 / 100

OFFICIAL VALIDATION

Results

100	100	100	0
Pre-twist	Post-twist routine	Post-twist demanding	Unsafe dispatches

- Leaderboard so far: 20 + 30 = 50 of 50 points
- Final private test (50%) runs our nominated version automatically

THANK YOU

What we would do next

- Learn how reliable stock records are while a batch runs
- Smarter multi-step planning when the budget is very tight
- Paid model tiers for higher rate limits at scale
- A live dashboard showing each decision and why it was made

Thank you - questions welcome